

Test coverage

Our Test coverage is 100% for the entire code (besides main). This coverage level indicates that every line of code, conditional branch, and function has been executed at least once during testing, ensuring comprehensive validation of functionality and behavior.

A screenshot of a terminal window showing a coverage report. At the top, it says 'Coverage report: 100%'. Below this are three buttons: 'Files', 'Functions', and 'Classes'. Underneath the buttons, it says 'coverage.py v7.6.4, created at 2024-11-11 00:34 -0500'. The main part of the report is a table with five columns: 'File', 'statements', 'missing', 'excluded', and 'coverage'. The table has three rows: 'Flight_booking\flightbook.py' with 217 statements, 0 missing, 3 excluded, and 100% coverage; 'Flight_booking\test_app.py' with 460 statements, 0 missing, 0 excluded, and 100% coverage; and a 'Total' row with 677 statements, 0 missing, 3 excluded, and 100% coverage. At the bottom, it repeats 'coverage.py v7.6.4, created at 2024-11-11 00:34 -0500'.

File	statements	missing	excluded	coverage ▾
Flight_booking\flightbook.py	217	0	3	100%
Flight_booking\test_app.py	460	0	0	100%
Total	677	0	3	100%

The `if __name__ == "__main__":` block, typically located at the end of the script, serves as the entry point for running the application. When the script is executed directly, it is responsible for starting the server. However, this block is unnecessary in testing environments because we don't need to start the server.

To exclude this block from the coverage report, we applied `# pragma: no cover`. This directive informs the coverage tool to ignore the specified lines, preventing us from directly testing the main block.

It allows us to achieve 100% meaningful coverage by focusing on the core functionality of the code while avoiding unnecessary execution of server startup code in the testing environment.